

Chapter 2

Variational Autoencoders

Exam Study Guide, Worked Concepts, Sample Questions, and

Prepared from	uploaded course materials
Main source	Chapter_2_Variational_Auto_Encoders.ipynb
Exam source	Chapter_2_Summary_and Sample_Questions.pdf
Practice source	Chapter_2_Flashcards.pdf

How to use it

This guide focuses on why VAEs are needed, the probabilistic model, marginal likelihood, ELBO, Jensen, reparameterization, sampling, latent geometry, β -VAE, CVAE, and failure modes.

Study order: First read the concept sections, then solve the five sample exam questions, then test yourself with the flashcards.

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- 4. Marginal likelihood and why direct optimization is hard
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- 11. β -VAE, CVAE, and failure modes
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Most useful mental shortcut

For every VAE question, identify whether it asks about the generative model, inference model, ELBO objective, or latent-space behavior.

1. Quick exam map

The chapter is about why a standard autoencoder is not enough as a generator, and how a VAE fixes this using a probabilistic latent-variable model.

Input x

a data example, such as an image Encoder $q\phi(z|x)$ outputs mean and variance, not one fixed code

Sample z

use reparameterization: $z = \mu + \sigma \cdot \epsilon$ Decoder $p\theta(x|z)$ outputs a probability distribution

Loss

reconstruction plus KL regularization Figure: the main method pipeline for this chapter.

Exam topic	What you must be able to say
Why VAE?	A deterministic AE reconstructs but does not define a principled probability
model $p\theta(x)$. Its latent space may contain holes.	Generative story
Sample z from $p(z)$, then generate x from $p\theta(x z)$.	Marginal likelihood

Exam topic	What you must be able to say
$p\theta(x) = \int p\theta(x z)p(z)dz$ is usually intractable.	Approximate posterior
Use $q\phi(z x)$ because the true posterior depends on intractable $p\theta(x)$.	ELBO
A tractable lower bound on $\log p\theta(x)$, made of reconstruction likelihood minus KL regularization.	Reparameterization
Moves randomness into ϵ so gradients can pass through z .	Disentanglement

Standard VAE does not guarantee factor separation; β -VAE and CVAE add useful biases.

2. Zero-prior explanation: understand the

idea before the equations This section assumes you have not read the lecture yet. The goal is to build intuition first, then connect the intuition to the equations and exam language.

Deterministic autoencoder vs. VAE A deterministic autoencoder maps x to one code z and reconstructs x . It can compress and denoise, but it does not define a full probability model. A VAE changes the encoder output into a distribution, making the latent space smoother and sampleable.

The probabilistic model behind a VAE

A VAE assumes each data point x was generated from an unobserved latent variable z . The prior $p(z)$ tells where latent codes usually live. The decoder $p\theta(x|z)$ explains how z produces data. The encoder $q\phi(z|x)$ guesses which z values explain an observed x .

Marginal likelihood and why it is hard

Maximum likelihood would maximize $\log p\theta(x)$, but $p\theta(x) = \int p\theta(x|z)p(z)dz$ requires integrating over all possible latent explanations. With high-dimensional z and neural decoders, this integral usually has no closed form.

Memory line

An AE learns reconstruction; a VAE learns a probabilistic latent-variable generative model.

3. Core concepts and worked explanations

Variational inference and approximate posterior

Since the true posterior $p\theta(z|x)$ uses the intractable $p\theta(x)$, the VAE introduces $q\phi(z|x)$. It is not assumed to be perfect; it is a tractable approximation that can be optimized.

ELBO and Jensen's intuition The ELBO is a lower bound on $\log p\theta(x)$. It rewards reconstruction while penalizing $q\phi(z|x)$ when it drifts too far from the prior. Jensen's inequality is the key step that turns log of an integral into an expectation of a log.

Reparameterization trick

Sampling directly from $q\phi(z|x)$ makes gradients difficult. Reparameterization writes $z = \mu\phi(x) + \sigma\phi(x)\epsilon$ with ϵ from a fixed standard normal. The randomness is separated from learnable parameters, so backpropagation works.

Sampling, geometry, and failure modes A useful latent space supports interpolation and prior sampling. Posterior collapse happens when $q\phi(z|x) \approx p(z)$, so z stops carrying input-specific information. β -VAE and CVAE modify the objective or conditioning to add structure.

4. Method explanation, worked example,

and application

Step	Method explanation
Encode distribution	Output μ and $\log \sigma^2$ for $q\phi(z x)$.
Sample z	Use $z = \mu + \sigma \cdot \epsilon$ so gradients pass through the sample.
Decode likelihood	Use $p\theta(x z)$ to reconstruct or generate data.
Optimize ELBO	Balance reconstruction likelihood and KL regularization.

Worked example

Example: for MNIST-like digits, the encoder maps each digit image to a small Gaussian region.

Nearby z values should decode into similar digits. Sampling z from $N(0, I)$ should produce plausible digits if the latent space is well organized.

Applications and usage

Usage: VAEs are used for representation learning, interpolation, anomaly detection, compression, controlled latent manipulation, and as conceptual foundations for latent generative models.

5. Formula / definition cheat sheet

Read formulas as compact sentences. Do not memorize symbols blindly; use the symbol key table after the formula table to understand what each part represents.

Formula	Interpretation
$p(z) = N(0, I)$ Standard latent prior.	$p\theta(x, z) = p\theta(x z)p(z)$ Joint generative model.
$p\theta(x) = \int p\theta(x z)p(z)dz$ Marginal likelihood; hard to compute directly.	$q\phi(z x) = N(\mu\phi(x), \text{diag}(\sigma\phi^2(x)))$ Approximate posterior / encoder.
$\mathcal{L}_{\text{ELBO}} = E_q[\log p\theta(x z)] - \text{KL}(q\phi(z x) p(z))$ Objective to maximize.	$\text{Loss} = -E_q[\log p\theta(x z)] + \text{KL}(q\phi(z x) p(z))$ Negative ELBO minimized in practice.
$z = \mu\phi(x) + \sigma\phi(x)\epsilon, \epsilon \sim N(0, I)$ Reparameterized sample.	$\mathcal{L}_\beta = E_q[\log p\theta(x z)] - \beta \text{KL}(q\phi(z x) p(z))$ β -VAE objective.

6. Equation symbol key table

This table explains the main symbols used in the equations. When studying, cover the meaning column and check whether you can explain each symbol clearly.

Symbol	Meaning	Plain-English reading
x	Observed data	The image or example being modeled.
z	Latent variable	Hidden code or cause behind x .
$p(z)$	Prior	Where z comes from before seeing x .
$p_{\theta}(x z)$	Decoder likelihood	How likely x is if latent code z is given.
$q\phi(z x)$	Approximate posterior	Encoder distribution over likely z values
after seeing x .	θ	Decoder parameters
Weights of the generative model.	ϕ	Encoder parameters
Weights of the inference model.	μ, σ	Mean and standard deviation
Parameters of the Gaussian encoder	output.	ϵ
Fixed noise sample	Randomness used in	reparameterization.
KL	KL divergence	Penalty for mismatch between $q\phi(z x)$ and $p(z)$.

7. Main code prompt / implementation

skeleton This is not meant to replace the lecture code. It is a compact prompt showing the computational pattern you should recognize in homework and exams.

```
# VAE training prompt
mu, logvar = encoder(x)
std = exp(0.5 * logvar)
eps = normal_like(std)
z = mu + std * eps
x_logits = decoder(z)
recon_loss = reconstruction_loss(x_logits, x)
kl = -0.5 * sum(1 + logvar - mu**2 - exp(logvar))
loss = recon_loss + kl
```

How to explain the code in words

Start by identifying the input, the model output, the objective being minimized, and the update rule. Then connect the code back to the formulas and the method table.

8. Common exam traps

Trap	Correct way to think
Saying the encoder generates	data The decoder/generative model generates x from z ; the encoder performs inference.
Forgetting the KL sign	ELBO maximization subtracts KL; negative-ELBO minimization adds KL.
Thinking disentanglement is	automatic It requires additional bias, supervision, or structure.

9. Exam-style sample questions with

answers These are written in the style of conceptual exam answers. Use them to practice explaining, not only recognizing terms.

Question 1

Why is the ELBO called a lower bound?

Sample answer

Because $\log p_{\theta}(x) = \text{ELBO} + \text{KL}(q_{\phi}(z|x) || p_{\theta}(z|x))$, and KL is nonnegative, so $\text{ELBO} \leq \log p_{\theta}(x)$.

Question 2

Explain the two ELBO terms.

Sample answer

The reconstruction term rewards explaining x ; the KL term keeps the approximate posterior close to the prior.

Question 3

Why is reparameterization more than coding convenience?

Sample answer

It makes stochastic sampling differentiable with respect to encoder outputs, enabling backpropagation.

Question 4

Why does a standard VAE not guarantee disentanglement?

Sample answer

The objective does not force each latent coordinate to represent one semantic factor.

Question 5

What does $q_{\phi}(z|x) \approx p(z)$ for every x suggest?

Sample answer

Posterior collapse: z carries little input-specific information and the decoder may ignore the latent code.

10. Flashcards for rapid review

Use these for active recall. Cover the answer column and try to answer before checking.

Card 1

Q: What does the VAE decoder define?

A: $p_{\theta}(x|z)$, a probability distribution over x .

Card 2

Q: What is the VAE prior?

A: Usually $p(z)=N(0,I)$.

Card 3

Q: What is $q\phi(z|x)$?

A: Approximate posterior implemented by the encoder.

Card 4

Q: Why is $p_{\theta}(x)$ hard?

A: It integrates over all latent z values.

Card 5

Q: What does KL regularize?

A: The approximate posterior toward the prior.

Card 6

Q: What is reparameterization?

A: Writing z as $\mu+\sigma\cdot\epsilon$.

Card 7

Q: What is posterior collapse?

A: $q\phi(z|x)\approx p(z)$, so z is uninformative.

Card 8

Q: What does β -VAE change?

A: It increases the KL weight.

Card 9

Q: What does CVAE condition on?

A: Known side information y .

Card 10

Q: Why can AE sampling fail?

A: Latent space may contain holes.

Card 11

Q: What must you remember about Why VAE??

A: A deterministic AE reconstructs but does not define a principled probability model $p_{\theta}(x)$. Its latent space may contain holes.

Card 12

Q: What must you remember about Generative story?

A: Sample z from $p(z)$, then generate x from $p_{\theta}(x|z)$.

Card 13

Q: What must you remember about Marginal likelihood?

A: $p_{\theta}(x)=\int p_{\theta}(x|z)p(z)dz$ is usually intractable.

Card 14

Q: What must you remember about Approximate posterior?

A: Use $q\phi(z|x)$ because the true posterior depends on intractable $p\theta(x)$.

Card 15

Q: What must you remember about ELBO?

A: A tractable lower bound on $\log p\theta(x)$, made of reconstruction likelihood minus KL regularization.

Card 16

Q: What must you remember about Reparameterization?

A: Moves randomness into ϵ so gradients can pass through z .

Card 17

Q: What must you remember about Disentanglement?

A: Standard VAE does not guarantee factor separation; β -VAE and CVAE add useful biases.

Card 18

Q: What does this formula represent: $p(z)=N(0,I)$?

A: Standard latent prior.

Card 19

Q: What does this formula represent:

$p\theta(x,z)=p\theta(x|z)p(z)$?

A: Joint generative model.

Card 20

Q: What does this formula represent:

$p\theta(x)=\int p\theta(x|z)p(z)dz$?

A: Marginal likelihood; hard to compute directly.

Card 21

Q: What does this formula represent:

$q\phi(z|x)=N(\mu\phi(x), \text{diag}(\sigma\phi^2(x)))$?

A: Approximate posterior / encoder.

Card 22

Q: What does this formula represent:

$\mathcal{L}_{ELBO}=E_q[\log p\theta(x|z)] - KL(q\phi(z|x)||p(z))$?

A: Objective to maximize.

Card 23

Q: What does this formula represent:

$\text{Loss}=-E_q[\log p\theta(x|z)] + KL(q\phi(z|x)||p(z))$?

A: Negative ELBO minimized in practice.

Card 24

Q: What does this formula represent:

$z=\mu\phi(x)+\sigma\phi(x)\epsilon, \epsilon\sim N(0,I)$?

A: Reparameterized sample.

Card 25

Q: What does this formula represent: $\mathcal{L}_\beta=E_q[\log p\theta(x|z)] - \beta KL(q\phi(z|x)||p(z))$?

A: β -VAE objective.

Card 26

Q: What does x mean?

A: Observed data. The image or example being modeled.

Card 27

Q: What does z mean?

A: Latent variable. Hidden code or cause behind x .

Card 28

Q: What does $p(z)$ mean?

A: Prior. Where z comes from before seeing x .

Card 29

Q: What does $p_\theta(x|z)$ mean?

A: Decoder likelihood. How likely x is if latent code z is given.

Card 30

Q: What does $q_\phi(z|x)$ mean?

A: Approximate posterior. Encoder distribution over likely z values after seeing x .

Card 31

Q: What does θ mean?

A: Decoder parameters. Weights of the generative model.

Card 32

Q: What does ϕ mean?

A: Encoder parameters. Weights of the inference model.

Card 33

Q: What does μ, σ mean?

A: Mean and standard deviation. Parameters of the Gaussian encoder output.

Card 34

Q: What does ϵ mean?

A: Fixed noise sample. Randomness used in reparameterization.

Card 35

Q: What does KL mean?

A: KL divergence. Penalty for mismatch between $q_\phi(z|x)$ and $p(z)$.

Card 36

Q: What is the key idea of Deterministic autoencoder vs. VAE?

A: A deterministic autoencoder maps x to one code z and reconstructs x . It can compress and denoise, but it does not define a full probability model. A VAE changes the encoder output into a distribution, making the latent.

Card 37

Q: What is the key idea of The probabilistic model behind a VAE?

A: A VAE assumes each data point x was generated from an unobserved latent variable z .

The prior $p(z)$ tells where latent codes usually live. The decoder $p_\theta(x|z)$ explains how z produces data. The encoder $q_\phi(z|x)$ guesses.

Card 38

Q: What is the key idea of Marginal likelihood and why it is hard?

A: Maximum likelihood would maximize $\log p_\theta(x)$, but $p_\theta(x) = \int p_\theta(x|z)p(z)dz$ requires integrating over all possible latent explanations. With high-dimensional z and neural decoders, this integral usually has no closed

Card 39

Q: What is the key idea of Variational inference and approximate posterior?

A: Since the true posterior $p_\theta(z|x)$ uses the intractable $p_\theta(x)$, the VAE introduces $q\phi(z|x)$. It is not assumed to be perfect; it is a tractable approximation that can be

Card 40

Q: What is the key idea of ELBO and Jensen intuition?

A: The ELBO is a lower bound on $\log p_\theta(x)$. It rewards reconstruction while penalizing $q\phi(z|x)$ when it drifts too far from the prior. Jensen inequality is the key step that turns log of an integral into an expectation of a.

Card 41

Q: What is the key idea of Reparameterization trick?

A: Sampling directly from $q\phi(z|x)$ makes gradients difficult. Reparameterization writes $z = \mu\phi(x) + \sigma\phi(x)\epsilon$ with ϵ from a fixed standard normal. The randomness is separated from learnable parameters, so backpropagation

Card 42

Q: What is the key idea of Sampling, geometry, and failure modes?

A: A useful latent space supports interpolation and prior sampling. Posterior collapse happens when $q\phi(z|x) \approx p(z)$, so z stops carrying input-specific information. β -VAE and CVAE modify the objective or conditioning to add.

Card 43

Q: What happens in the step: Encode distribution?

A: Output μ and $\log \sigma^2$ for $q\phi(z|x)$.

Card 44

Q: What happens in the step: Sample z ?

A: Use $z = \mu + \sigma \cdot \epsilon$ so gradients pass through the sample.

Card 45

Q: What happens in the step: Decode likelihood?

A: Use $p_\theta(x|z)$ to reconstruct or generate data.

Card 46

Q: What happens in the step: Optimize ELBO?

A: Balance reconstruction likelihood and KL regularization.

Card 47

Q: Why is a VAE needed?

A: AE reconstructs but lacks a principled sampleable probability model.

Card 48

Q: What is $p_\theta(x)$?

A: The marginal likelihood integrating out z .

Card 49

Q: Why introduce $q\phi(z|x)$?

A: The true posterior is intractable.

Card 50

Q: Where does Jensen appear?

A: It creates the ELBO lower bound.

Card 51

Q: What causes posterior collapse?

A: Overly powerful decoder or strong KL pressure.

Card 52

Q: Why is the ELBO called a lower bound?

A: Because $\log p_{\theta}(x) = \text{ELBO} + \text{KL}(q\phi(z|x) || p_{\theta}(z|x))$, and KL is nonnegative, so $\text{ELBO} \leq \log p_{\theta}(x)$.

Card 53

Q: Explain the two ELBO terms.

A: The reconstruction term rewards explaining x ; the KL term keeps the approximate posterior close to the prior.

Card 54

Q: Why is reparameterization more than coding convenience?

A: It makes stochastic sampling differentiable with respect to encoder outputs, enabling backpropagation.

Card 55

Q: Why does a standard VAE not guarantee disentanglement?

A: The objective does not force each latent coordinate to represent one semantic factor.

Explanation

Chapter 2: Variational Autoencoders

How this section is organized

Each item below is a heading taken from the corresponding Jupyter notebook in the same order it appears there. The explanation beside it tells you what that notebook section is trying to teach and why it matters for the review.

Source notebook

Chapter_2_Variational_Auto_Encoders.ipynb

Headings included

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Notebook headings with explanation

1. Chapter 2 — Variational Autoencoders	This chapter is an introduction to Variational Autoencoders
2. Learning Objectives	By the end of this chapter, you should be able to: 1. Explain variational autoencoders 2.
3. 1 Deterministic Autoencoder vs. Variational AutoEncoder	A deterministic autoencoder maps: - x to z to $\hat{x}$ A VAE maps: - x to $(\mu(x), \log \sigma^2(x))$ - sample $z \sim q_\phi(z x)$ using reparameterization - decode z to $\hat{x}$ using a probabilistic decoder $p_\theta(x z)$
4. 2 Why go beyond a deterministic Autoencoder?	A deterministic AE learns: x is encoded into z is decoded into $\hat{x}$ Great for compression/reconstruction, but not a principled generator: - no explicit model for $p_\theta(x)$ - latent space can be irregular ("holes") - sampling random z (e.g., from $N(0, I)$) often decodes to nonsense A VAE makes z probabilistic and regularized, so sampling is meaningful. A deterministic autoencoder (AE) learns a *single* code for each input: x is encoded into f_ϕ z is decoded into g_θ $\hat{x}$ and typically.
5. 3 VAE basic idea	Prior on latent variables We choose a simple prior: $p(z) = N(0, I)$ Decoder = generative model The decoder defines a likelihood where we need to estimate θ : $p_\theta(x z)$ - For images in $[0, 1]$, a common choice is Bernoulli per pixel (BCE loss). - For real-valued vectors, a common choice is Gaussian (MSE loss corresponds to fixed-variance Gaussian). Encoder = variational posterior We approximate the intractable true posterior $p_\theta(z x)$ with: $q_\phi(z x) = N(\mu_\phi(x), \text{diag}(\sigma_\phi^2(x)))$
6. 4 Maximizing Data Likelihood	When the model is trained, we will give the decoder sampled z as input and it will generate x (e.g., an image) using the decoder. Sample a latent variable: $z \sim p(z)$ 2. The notebook also includes 1 nearby code cell for implementation, demonstration, or visualization of this heading.
7. 5 1D Simple Latent-Variable Model	We define: - Prior: $z \sim N(0, 1)$ - Decoder: $x \mid z \sim N(f(z), \sigma_x^2)$ with a nonlinear mean function (like a tiny "neural decoder"). In reality, this will be a neural network: $f(z) = z^3$ Even in 1D, this makes the marginal: $p(x) = \int N(x; f(z), \sigma_x^2) N(z; 0, 1) dz$ nontrivial, but we *can* compute it numerically with a dense grid. The notebook also includes 3 nearby code cells for implementation, demonstration, or visualization of this heading.
8. 6 Monte Carlo estimate of $\log p(x)$ (and why it can be noisy)	A naive Monte Carlo estimator for $p(x)$ is: $p(x) \approx \frac{1}{K} \sum_{k=1}^K p(x \mid z_k)$, $z_k \sim p(z)$ Then: $\log p(x) \approx \log \left(\frac{1}{K} \sum_{k=1}^K p(x \mid z_k) \right)$ Two practical issues: 1. Most prior samples $z_k \sim p(z)$ may land in regions where $p(x \mid z_k)$ is tiny. The notebook also includes 1 nearby code cell for implementation, demonstration, or visualization of this heading.

9. 7 The Evidence Lower Bound (ELBO): a tractable lower bound	Since $\log p_{\theta}(x)$ is generally not tractable, we optimize a lower bound that is differentiable and efficient. The idea is that if we maximize the lower bound, in some sense we are maximizing $\log p_{\theta}(x)$.
10. Integrand and the True Posterior	For a fixed x_0 , the true posterior satisfies: $p(z \mid x_0) \propto p(x_0 \mid z)p(z)$. So the *shape* of the posterior is exactly the shape of the integrand. The notebook also includes 1 nearby code cell for implementation, demonstration, or visualization of this heading.
11. Fix a Specific Observation	Now choose a specific data point: $x = x_0$. Then: $p(z \mid x_0) = \frac{p(x_0 \mid z)p(z)}{\int p(x_0 \mid z')p(z')dz'}$.
12. Recognize the Proportional Form	The denominator does not depend on z . It is simply a normalizing constant.
13. Summary	For a fixed observation x_0 : - Values of z that make x_0 likely (high $p(x_0 \mid z)$) - And are themselves likely under the prior (high $p(z)$) will have high posterior probability. The denominator simply rescales the result so it integrates to 1.
14. 8 ELBO Estimator for a Gaussian Variational Family	We restrict our approximation to: $q(z \mid x_0) = \mathcal{N}(\mu_q, \sigma_q^2)$. The ELBO can be written as: $\text{ELBO}(x_0) = \mathbb{E}_q[\log p(x_0 \mid z) + \log p(z) - \log q(z \mid x_0)]$. The notebook also includes 3 nearby code cells for implementation, demonstration, or visualization of this heading.
15. 9 True Posterior vs. Learned Gaussian $q(z \mid x_0)$	Now we overlay: - True posterior $p(z \mid x_0)$ (from numerical normalization) - Best Gaussian approximation $q(z \mid x_0)$ (from ELBO maximization) If the true posterior is not Gaussian, then even the best Gaussian cannot match it perfectly. That mismatch corresponds to: $\log p(x_0) - \text{ELBO}(x_0) = \text{KL}(q(z \mid x_0) \parallel p(z \mid x_0)) \geq 0$. The notebook also includes 1 nearby code cell for implementation, demonstration, or visualization of this heading.
16. 10 Encoder and decoder math in one place	Encoder (learned) Encoder network parameters: ϕ Outputs: $\mu_{\phi}(x)$, $\log \sigma_{\phi}^2(x)$ Defines: $q_{\phi}(z \mid x) = \mathcal{N}(\mu_{\phi}(x), \text{diag}(\sigma_{\phi}^2(x)))$ Reparameterization $\epsilon \sim \mathcal{N}(0, I)$, $z = \mu_{\phi}(x) + \sigma_{\phi}(x) \odot \epsilon$ Decoder (learned) Decoder network parameters: θ Defines likelihood: $p_{\theta}(x \mid z)$ Common cases: - Bernoulli decoder (MNIST-like): $p_{\theta}(x \mid z) = \prod_i \text{Bernoulli}(x_i; \hat{z}_i)$ Training uses BCE. - Gaussian decoder (real-valued): $p_{\theta}(x \mid z) = \mathcal{N}(x; \mu_{\theta}(z), \sigma_{\theta}^2)$ Training.
17. 11 The reparameterization trick — theory, motivation, and why it works	The core problem: gradients through randomness In a VAE, the encoder defines a distribution, not a single point and therefore we cannot back-propagate through it directly: $q_{\phi}(z \mid x) = \mathcal{N}(\mu_{\phi}(x), \sigma_{\phi}^2(x))$. The ELBO contains an expectation over this distribution: $\mathbb{E}_q[\log p_{\theta}(x \mid z)]$.
18. Examples	Now we show some examples of how AE fails and VAE succeeds in learning the distribution. We start with generating 1D dataset where we are trying to learn a simple mixture of Gaussians. The notebook also includes 1 nearby code cell for implementation, demonstration, or visualization of this heading.
19. 12 Sampling fails in AE	When we reconstruct, we can get a good reconstruction. However, if we sample the z , the output is not correct. The notebook also includes 1 nearby code cell for implementation, demonstration, or visualization of this heading.
20. 13 VAE implementation (Gaussian latent)	VAE Performs Better Unlike AE, the VAE performs better at modelling the sampling distribution. Applying VAE to generate MNIST The VAE learned a smooth, Gaussian-aligned latent manifold for MNIST, but the representation is entangled and dominated by a single principal direction rather than clean independent factors. The notebook also includes 5 nearby code cells for implementation, demonstration, or visualization of this heading.

21. 14 Entanglement and Disentanglement in Latent Variable Models	What do we mean by *entanglement*? In a latent-variable model (such as a VAE), we say the representation is entangled when: > Each latent dimension mixes multiple underlying factors of variation in the data.
22. 15 Inductive Bias and why disentanglement does not come for free	An inductive bias is any assumption we build into a model that restricts the set of solutions it can learn. More formally: > Inductive bias is the set of preferences a learning algorithm uses to choose one solution over many equally valid solutions that fit the data.
23. 16 How β-VAE Addresses the Entanglement Problem	The core problem β -VAE is trying to fix A standard VAE has too much freedom in how it uses the latent space. As long as it: - reconstructs the data well, and - keeps the latent distribution roughly Gaussian, it can mix multiple factors together inside each latent dimension.
24. 17 An example to demonstrate disentanglement using β-VAE	This example is designed to show how a β -VAE encourages disentanglement in a controlled 1D setting. Even though the observed data is 1-dimensional, we intentionally construct it to depend on two independent underlying factors: $x = s \cdot a + \text{noise}$ where: - $s \in \{-1, +1\}$ is a discrete sign factor - $a \in [0.2, 2.0]$ is a continuous amplitude factor So although we observe only a scalar x , it is generated from two independent generative causes. The notebook also includes 1 nearby code cell for implementation, demonstration, or visualization of this heading.
25. 18 How Conditional VAEs (CVAEs) Help with the Entanglement Problem	The core idea A Conditional Variational Autoencoder (CVAE) reduces entanglement by telling the model some of the answers upfront. Instead of forcing the latent space to explain *everything* about the data, a CVAE says: > "Here is a known factor (like a label).
26. 19 Conditional VAE (CVAE): why conditioning makes sense	Suppose each data point has a label or condition y (class, attribute, text prompt, etc.). Goal Instead of modeling $p(x)$, we want to model: $p(x y)$ So we can generate samples controlled by y .
27. 20 CVAE ELBO	We maximize conditional likelihood: $\log p_{\theta}(x y) = \log \int p_{\theta}(x z,y)p(z)dz$ Introduce $q_{\phi}(z x,y)$ and apply Jensen: $\log p_{\theta}(x y) \geq \mathbb{E}_{q_{\phi}(z x,y)}[\log p_{\theta}(x z,y)] - \text{KL}(q_{\phi}(z x,y) \ p(z))$ So the CVAE ELBO is: $L^{\text{cond}}_{\text{ELBO}}(x,y) = \mathbb{E}_{q_{\phi}(z x,y)}[\log p_{\theta}(x z,y)] - \text{KL}(q_{\phi}(z x,y) \ p(z))$ What changes vs VAE? - Reconstruction term becomes conditioned: $\log p_{\theta}(x z,y)$ - Posterior becomes conditioned: $q_{\phi}(z x,y)$ - (Often) the prior stays $p(z)$ Which parameters are.
28. 21 1D Conditional VAE and Disentanglement Example	Data Generating Process We construct synthetic data from two independent latent factors: $x = s \cdot a + \epsilon$ where: - $s \in \{-1, +1\}$ is a discrete sign factor - $a \in [0.2, 2.0]$ is a continuous amplitude factor - $\epsilon \sim \mathcal{N}(0, \sigma^2)$ is small Gaussian noise Although the observation $x \in \mathbb{R}$ is 1-dimensional, it depends on two independent causes. So ideally: $z \rightarrow a$ and $y \rightarrow s$ This creates structured separation of generative factors. The notebook also includes 1 nearby code cell for implementation, demonstration, or visualization of this heading.
29. 23 MNIST in Parallel: VAE vs β-VAE vs CVAE	Below is a parallel comparison showing how the *same MNIST task* is handled by four related models. Assume: - $x \in [0,1]^{784}$ is a flattened MNIST image - $y \in \{0, \dots, 9\}$ is the digit label - $\text{onehot}(y) \in \{0,1\}^{10}$ is its one-hot vector - $z \in \mathbb{R}^d_z$ is a latent vector - $E = e_1, \dots, e_K$ is a VQ-VAE codebook
30. Standard VAE (baseline)	This notebook heading introduces standard vae (baseline) and explains how it supports the chapter concepts, equations, method steps, or implementation workflow.
31. β-VAE (for disentanglement)	Goal on MNIST Same as VAE, but push the latent space to be simpler and more factorized so that: - different dimensions of z are more interpretable - style factors separate more cleanly (thickness, slant, etc.) What changes? Only the weighting on the KL term.

32. Conditional VAE (CVAE) (control digit identity)	Goal on MNIST Model: $p(x y)$ so you can generate a specific digit on command. This reduces “identity-style mixing” inside z . The notebook also includes 5 nearby code cells for implementation, demonstration, or visualization of this heading.
33. 25 Common Failure Modes in VAEs	This notebook heading introduces 25 common failure modes in vae's and explains how it supports the chapter concepts, equations, method steps, or implementation workflow.
34. Posterior collapse	Symptom: $\text{KL}(\text{bigl}(q(z \text{ mid } x), p(z) \text{ bigr}); \text{to}; 0)$ Latent variables are effectively ignored. The encoder learns to output almost the same distribution for every input: $q(z \text{ mid } x) \approx p(z)$ This means: - z carries little or no information about x - the decoder learns to reconstruct using its own parameters only In effect, the model becomes: > “A decoder pretending to be a generative model.” Why this happens - The decoder is too expressive - The KL penalty is applied too strongly or too early - It is *easier*.
35. Over-regularization (β too large)	Symptom: Reconstructions degrade, samples look empty or distorted What is happening? In a β -VAE, the objective is: $\mathbb{E}_q(z \text{ mid } x)[\log p(x \text{ mid } z)] - \beta \text{KL}(\text{bigl}(q(z \text{ mid } x), p(z) \text{ bigr})$ When β is too large: - matching the prior becomes more important than reconstruction - the latent space is forced to carry very little information Why this happens - β -VAE intentionally restricts information capacity - Pushing β too far removes essential details Why it's bad - Reconstructions become inaccurate -.
36. Blurry outputs (likelihood mismatch)	Symptom: Generated images look smooth, averaged, or fuzzy What is happening? Using Mean Squared Error corresponds to assuming: $p(x \text{ mid } z) = \mathcal{N}(\hat{x}(z), \sigma^2 I)$ This loss encourages predicting the mean of all plausible outputs.
37. ELBO tightness gap	Symptom: ELBO improves, but sample quality does not What is happening? The Evidence Lower Bound satisfies: $\log p(x) \geq \text{ELBO}(x)$ If the approximate posterior $q(z \text{ mid } x)$ is too simple: - it cannot match the true posterior - the bound becomes loose The model can improve the bound without learning a better generator.
38. 25 Prior vs Posterior in VAEs	Understanding prior and posterior is the key to understanding *why VAEs work* and *why they fail*. What problem are we trying to solve?
39. 26 Summary: Mathematical Goal of a VAE	The goal of a Variational Autoencoder is to learn a generative model: $p_{\theta}(x, z) = p_{\theta}(x \text{ mid } z), p(z)$ such that the marginal distribution $p_{\theta}(x) = \int p_{\theta}(x \text{ mid } z), p(z), dz$ matches the true data distribution: $p_{\text{data}}(x)$. The True Objective Ideally, we would like: $p_{\theta}(x) \approx p_{\text{data}}(x)$.